Sudhir Pol

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Summary

Machine Learning Engineer focused on LLM inference and serving efficiency. Writes CUDA attention kernels, quantizes models to 4 bit, and serves them on vLLM, with 3 years of production experience deploying transformer services on AWS at Adobe, S&P Global, and American Express. Publishes on quantization math and speculative decoding.

Inference Projects

FlashAttention V2 from Scratch | *CUDA C++, Nsight Compute* **Apr. 2026**

- Implemented the FlashAttention V1 and V2 forward pass in CUDA using tiled attention with online softmax, matching PyTorch scaled dot product attention output to 2e-6 in fp32
- Optimized the kernel 8.5 times in Nsight Compute by restructuring the tiling loop to remove redundant HBM traffic, eliminating shared memory bank conflicts, and parallelizing the softmax with warp shuffle reductions

QuantRoute: Quantization Aware Inference Router | *vLLM, llm-compressor, AWQ, GPTQ, A100* **Nov. 2025**

- Quantized Qwen2.5-7B to 4 bit with AWQ and GPTQ using llm-compressor and served multiple precision lanes on vLLM, reaching 2.2 times faster decode at 56% lower serving cost on financial tasks
- Built a learned difficulty classifier with an LLM as Judge escalation loop so only hard requests reach the expensive lane, using offline oracle analysis to isolate routing bottlenecks

Speculative Decoding from Scratch | *PyTorch, HuggingFace Transformers* **Oct. 2025**

- Implemented draft and verify speculative decoding end to end, including the accept reject sampling step and the adjusted residual distribution that preserves the target model output distribution
- Benchmarked across draft lengths and measured 3.4 times faster generation at gamma 4, 95 tokens per second against a 28 token per second baseline, with acceptance rate falling to 44% at gamma 8

Experience

Adobe **May 2025 - Aug. 2025**
Machine Learning Engineer Intern *San Jose, California*

- Benchmarked Gemini 2.5 Pro, GPT 4.1, and Claude 4.5 behind a single LiteLLM interface in an LLM as Judge evaluation harness, raising offline F1 score from 79% to 92% by rebuilding the scoring metrics
- Shipped a LangGraph agent over Model Context Protocol that cut an evaluation workflow from 48 hours to 30 minutes, with a traced human approval gate before any plan executed

S&P Global **Nov. 2022 - Aug. 2024**
Data Scientist II *Hyderabad, India*

- Owned a Llama 2 inference service on SageMaker and AWS Lambda end to end, from architecture through deployment and support, removing about 20 minutes of manual lookup per analyst task
- Fine tuned and deployed LayoutLMv3 for token classification on AWS EC2, and served a T5 question answering API over long financial filings in the same environment

American Express **Dec. 2021 - Nov. 2022**
Data Scientist I *Gurgaon, India*

- Built and containerized the VIBE call quality service with Docker, combining XGBoost with BERT based embeddings, which ran as the system of record for representative bonuses
- Deployed an LSTM model with Flask on AWS EC2 that mapped call transcripts to demand types, letting the business route high priority customer calls first

Writing

- The Math of AWQ: Protecting Salient Channels from the Inside Out *Jun. 2026*
- Demystifying GPTQ: From Lagrange Multipliers to Vectorized PyTorch *Jun. 2026*
- Speculative Decoding: Make LLM Inference Faster *Oct. 2025*

Skills

Languages: Python, C++, CUDA C++, SQL, Java
Inference and Optimization: Quantization (AWQ, GPTQ, 4 bit), Speculative Decoding, FlashAttention, Kernel Profiling, Throughput and Latency Benchmarking, Model Serving
Frameworks: PyTorch, vLLM, llm-compressor, HuggingFace Transformers, PEFT, LiteLLM, LangChain, FastAPI
Infrastructure: CUDA, Nsight Compute, NVIDIA A100, Docker, Kubernetes, AWS (SageMaker, EC2, Lambda), GitHub Actions, MLflow

Education

Indiana University **Aug. 2024 - May 2026**
Master of Science in Data Science *Bloomington, United States*

Rajarambapu Institute of Technology **Aug. 2017 - May 2021**
Bachelor of Technology in Electronics and Tele-Communications *Sangli, India*